glm-4.7-hf-model-card

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language:

- en
- zh library_name: transformers license: mit pipeline_tag: text-generation

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GLM-4.7



👋 Join our Discord community.
📖 Check out the GLM-4.7 technical blog, technical report(GLM-4.5).
📍 Use GLM-4.7 API services on Z.ai API Platform.
👉 One click to GLM-4.7.

Introduction

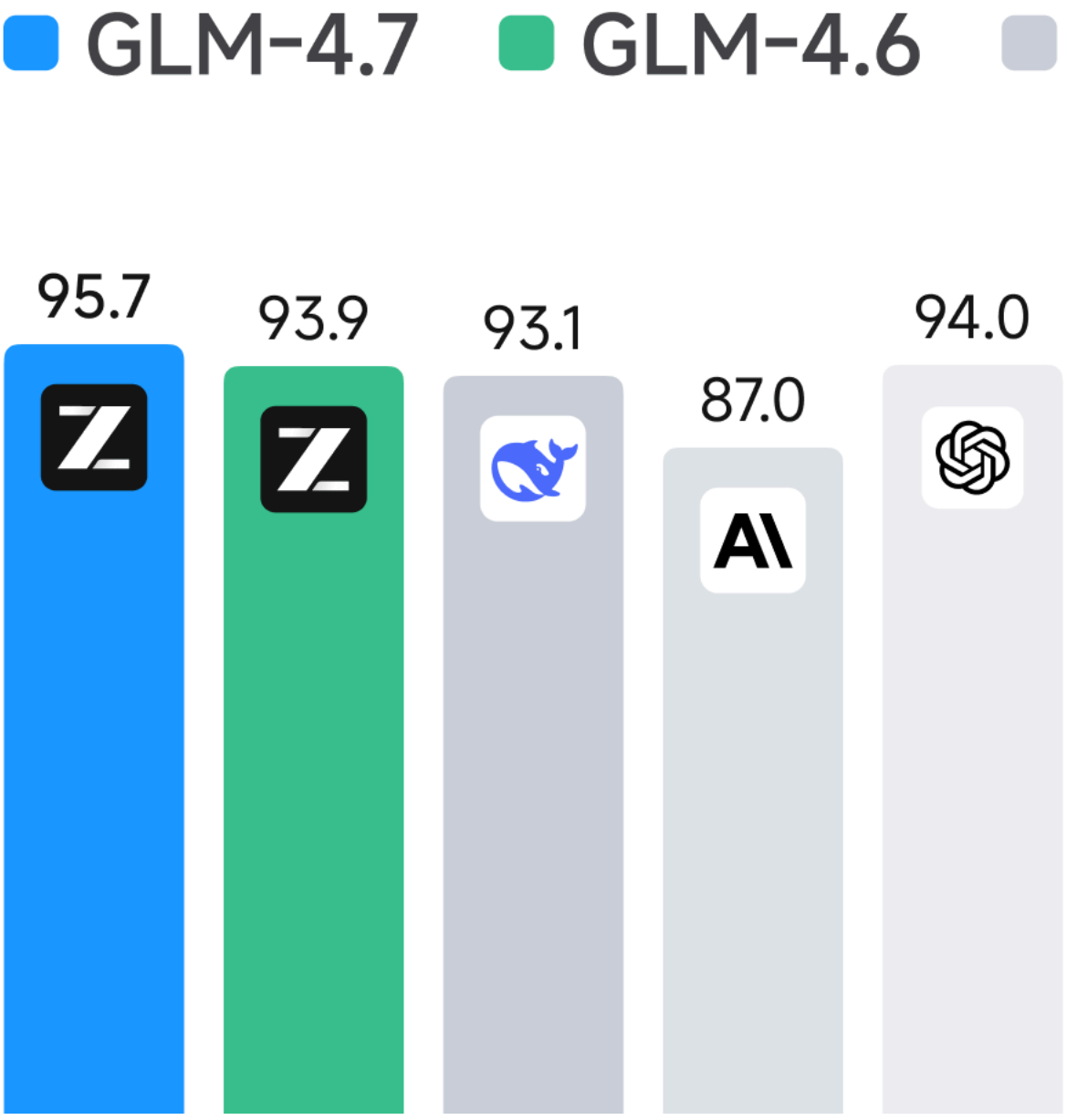
GLM-4.7, your new coding partner, is coming with the following features:

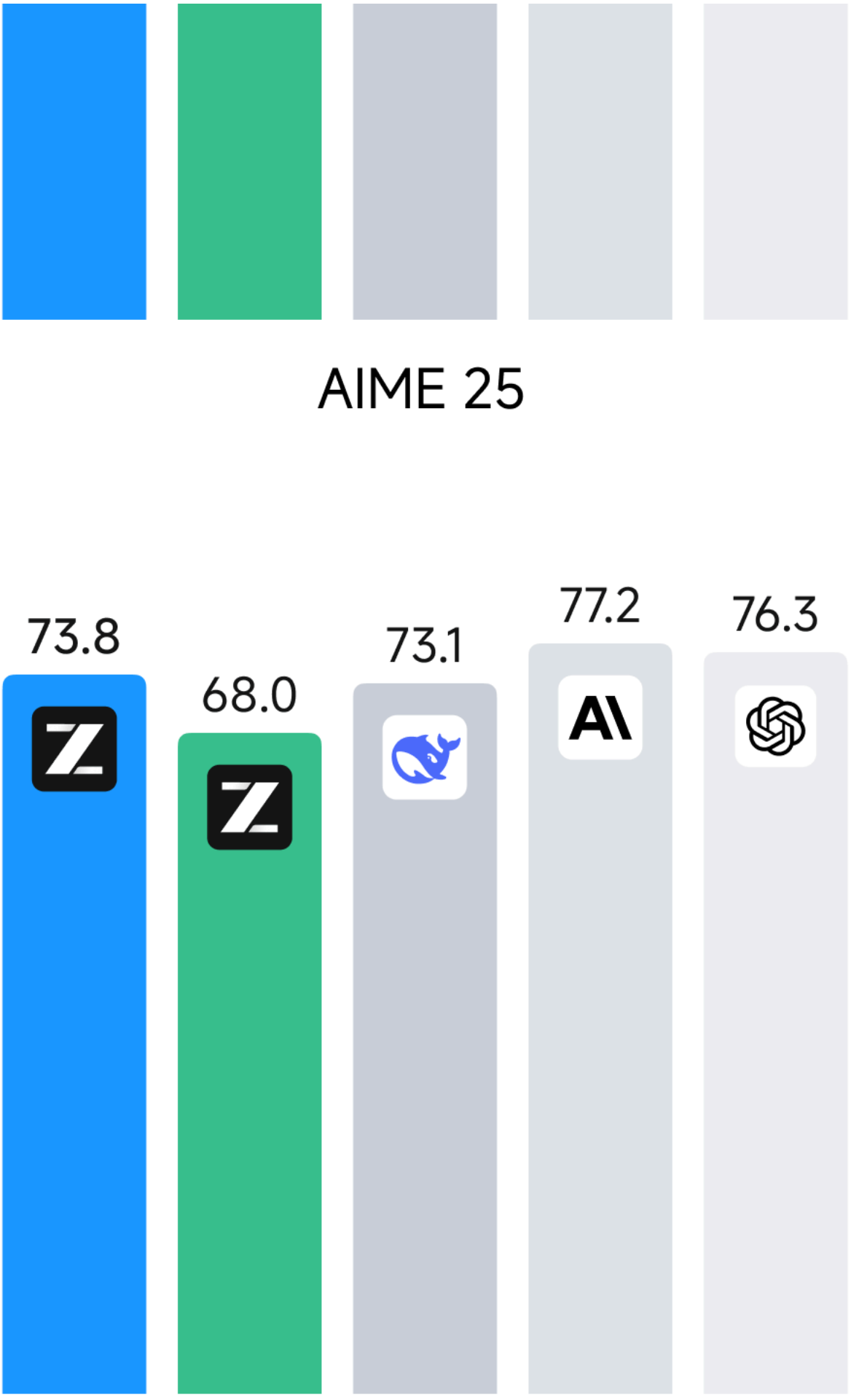
- **Core Coding**: GLM-4.7 brings clear gains, compared to its predecessor GLM-4.6, in multilingual agentic coding and terminal-based tasks, including (73.8%, +5.8%) on SWE-bench, (66.7%, +12.9%) on SWE-bench Multilingual, and (41%, +16.5%) on Terminal Bench 2.0. GLM-4.7 also supports thinking before acting, with significant improvements on complex tasks in mainstream agent frameworks such as Claude Code, Kilo Code, Cline, and Roo Code.
- **Vibe Coding**: GLM-4.7 takes a big step forward in improving UI quality. It produces cleaner, more modern webpages and generates better-looking slides with more accurate layout and sizing.
- **Tool Using**: GLM-4.7 achieves significantly improvements in Tool using. Significant better performances can be seen on benchmarks such as τ^2-Bench and on web browsing via BrowseComp.
- **Complex Reasoning**: GLM-4.7 delivers a substantial boost in mathematical and reasoning capabilities, achieving (42.8%, +12.4%) on the HLE (Humanity's Last Exam) benchmark compared to GLM-4.6.

You can also see significant improvements in many other scenarios such as chat, creative writing, and role-play scenario.

LLM Performanc

8 benchmarks: AIME 25, LiveC
(Evaluation results under 128K




AIME 25
73.8
68.0
73.1
77.2
76.3

SWE-bench Verified

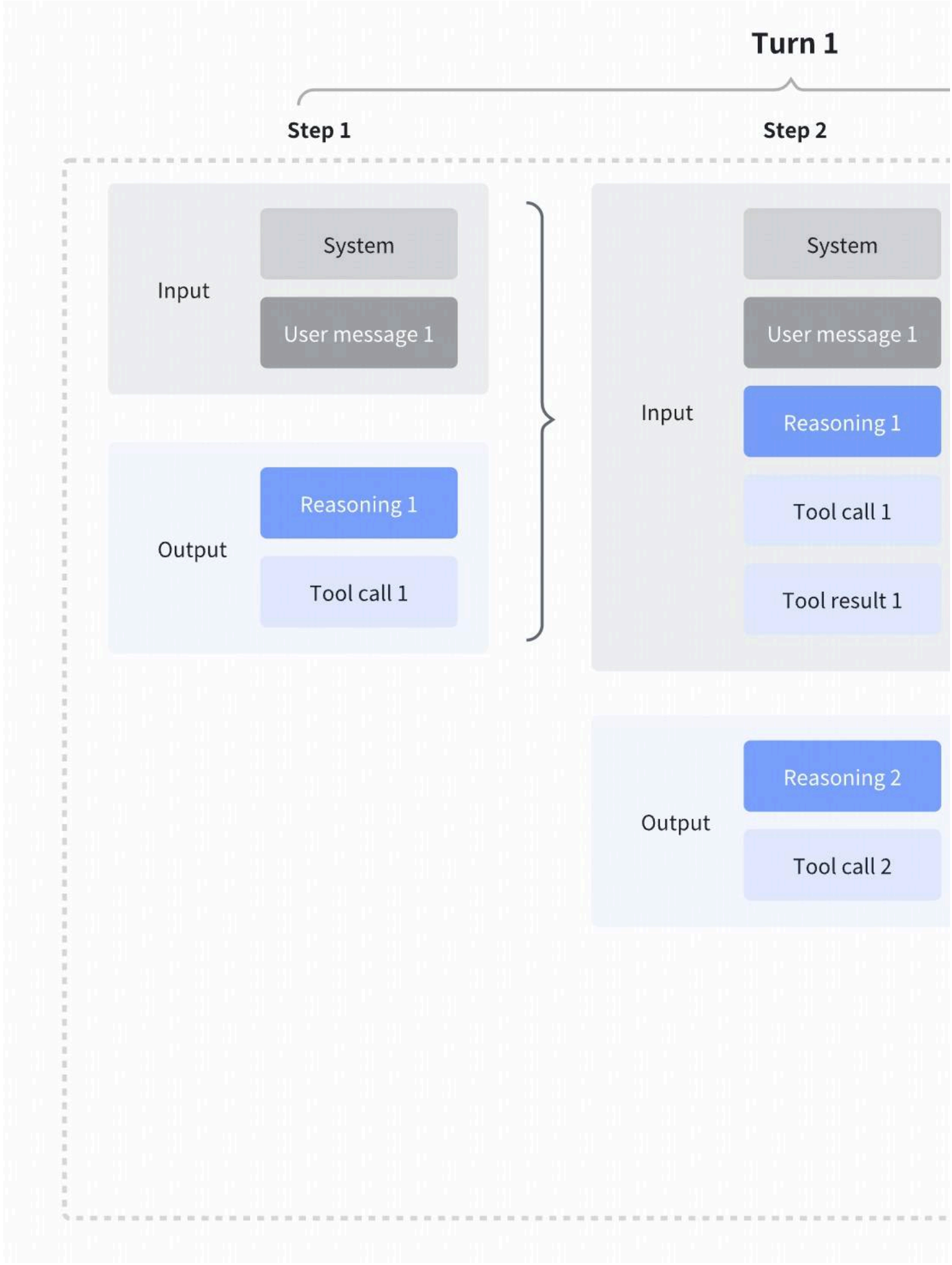
Performances on Benchmarks. More detailed comparisons of GLM-4.7 with other models GPT-5-High, GPT-5.1-High, Claude Sonnet 4.5, Gemini 3.0 Pro, DeepSeek-V3.2, Kimi K2 Thinking, on 17 benchmarks (including 8 reasoning, 5 coding, and 3 agents benchmarks) can be seen in the below table.

Benchmark	GLM-4.7	GLM-4.6	Kimi K2 Thinking	DeepSeek-V3.2	Gemini 3.0 Pro	Claude Sonnet 4.5	GPT-5-High	GPT-5.1-High
MMLU-Pro	84.3	83.2	84.6	85.0	90.1	88.2	87.5	87.0
GPQA-Diamond	85.7	81.0	84.5	82.4	91.9	83.4	85.7	88.1
HLE	24.8	17.2	23.9	25.1	37.5	13.7	26.3	25.7
HLE (w/ Tools)	42.8	30.4	44.9	40.8	45.8	32.0	35.2	42.7
AIME 2025	95.7	93.9	94.5	93.1	95.0	87.0	94.6	94.0
HMMT Feb. 2025	97.1	89.2	89.4	92.5	97.5	79.2	88.3	96.3
HMMT Nov. 2025	93.5	87.7	89.2	90.2	93.3	81.7	89.2	-
IMOAnswerBench	82.0	73.5	78.6	78.3	83.3	65.8	76.0	-
LiveCodeBench-v6	84.9	82.8	83.1	83.3	90.7	64.0	87.0	87.0
SWE-bench Verified	73.8	68.0	71.3	73.1	76.2	77.2	74.9	76.3
SWE-bench Multilingual	66.7	53.8	61.1	70.2	-	68.0	55.3	-
Terminal Bench Hard	33.3	23.6	30.6	35.4	39.0	33.3	30.5	43.0
Terminal Bench 2.0	41.0	24.5	35.7	46.4	54.2	42.8	35.2	47.6
BrowseComp	52.0	45.1	-	51.4	-	24.1	54.9	50.8
BrowseComp (w/ Context Manage)	67.5	57.5	60.2	67.6	59.2	-	-	-
BrowseComp-Zh	66.6	49.5	62.3	65.0	-	42.4	63.0	-
τ^2-Bench	87.4	75.2	74.3	85.3	90.7	87.2	82.4	82.7

Coding: AGI is a long journey, and benchmarks are only one way to evaluate performance. While the metrics provide necessary checkpoints, the most important thing is still how it *feels*. True intelligence isn't just about acing a test or processing data faster; ultimately, the success of AGI will be measured by how seamlessly it integrates into our lives-**"coding"** this time.

Getting started with GLM-4.7

Interleaved Thinking & Preserved Thinking


Turn 1
Step 1
Step 2
Input
System
User message 1
Output
Reasoning 1
Tool call 1
Input
System
User message 1
Reasoning 1
Tool call 1
Tool result 1
Output
Reasoning 2
Tool call 2

GLM-4.7 further enhances **Interleaved Thinking** (a feature introduced since GLM-4.5) and introduces **Preserved Thinking** and **Turn-level Thinking**. By thinking between actions and staying consistent across turns, it makes complex tasks more stable and more controllable:

- **Interleaved Thinking**: The model thinks before every response and tool calling, improving instruction following and the quality of generation.
- **Preserved Thinking**: In coding agent scenarios, the model automatically retains all thinking blocks across multi-turn conversations, reusing the existing reasoning instead of re-deriving from scratch. This reduces information loss and inconsistencies, and is well-suited for long-horizon, complex tasks.
- **Turn-level Thinking**: The model supports per-turn control over reasoning within a session—disable thinking for lightweight requests to reduce latency/cost, enable it for complex tasks to improve accuracy and stability.

More details: https://docs.z.ai/guides/capabilities/thinking-mode

Evaluation Parameters

Default Settings (Most Tasks)

- temperature: `1.0`
- top-p: `0.95`
- max new tokens: `131072`

For multi-turn agentic tasks (τ²-Bench and Terminal Bench 2), please turn on Preserved Thinking mode.

Terminal Bench, SWE Bench Verified

- temperature: `0.7`
- top-p: `1.0`
- max new tokens: `16384`

τ^2-Bench

- Temperature: `0`
- Max new tokens: `16384`

For τ^2-Bench evaluation, we added an additional prompt to the Retail and Telecom user interaction to avoid failure modes caused by users ending the interaction incorrectly. For the Airline domain, we applied the domain fixes as proposed in the Claude Opus 4.5 release report.

Serve GLM-4.7 Locally

For local deployment, GLM-4.7 supports inference frameworks including vLLM and SGLang. Comprehensive deployment instructions are available in the official Github repository.

vLLM and SGLang only support GLM-4.7 on their main branches. you can use their official docker images for inference.

vLLM

Using Docker as:

```
docker pull vllm/vllm-openai:nightly
```

or using pip (must use pypi.org as the index url):

```
pip install -U vllm --pre --index-url https://pypi.org/simple --extra-index-url https://wheels.vllm.ai/nightly
```

SGLang

Using Docker as:

```
docker pull lmsysorg/sglang:dev
```

or using pip install sglang from source.

transformers

using with transformers as `4.57.3` and then run:

```
import torch
from transformers import AutoModelForCausalLM, AutoTokenizer

MODEL_PATH = "zai-org/GLM-4.7"
```

```
messages = [{"role": "user", "content": "hello"}]
tokenizer = AutoTokenizer.from_pretrained(MODEL_PATH)
inputs = tokenizer.apply_chat_template(
    messages,
    tokenize=True,
    add_generation_prompt=True,
    return_dict=True,
    return_tensors="pt",
)
model = AutoModelForCausalLM.from_pretrained(
    pretrained_model_name_or_path=MODEL_PATH,
    torch_dtype=torch.bfloat16,
    device_map="auto",
)
inputs = inputs.to(model.device)
generated_ids = model.generate(**inputs, max_new_tokens=128, do_sample=False)
output_text = tokenizer.decode(generated_ids[0][inputs.input_ids.shape[1] :])
print(output_text)
```

vLLM

```
vllm serve zai-org/GLM-4.7-FP8 \
     --tensor-parallel-size 4 \
     --speculative-config.method mtp \
     --speculative-config.num_speculative_tokens 1 \
     --tool-call-parser glm47 \
     --reasoning-parser glm45 \
     --enable-auto-tool-choice \
     --served-model-name glm-4.7-fp8
```

SGLang

```
python3 -m sglang.launch_server \
  --model-path zai-org/GLM-4.7-FP8 \
  --tp-size 8 \
  --tool-call-parser glm47  \
  --reasoning-parser glm45 \
  --speculative-algorithm EAGLE \
  --speculative-num-steps 3 \
  --speculative-eagle-topk 1 \
  --speculative-num-draft-tokens 4 \
  --mem-fraction-static 0.8 \
  --served-model-name glm-4.7-fp8 \
  --host 0.0.0.0 \
  --port 8000
```

Parameter Instructions

- For agentic tasks of GLM-4.7, please turn on Preserved Thinking mode by adding the following config (only sglang support):

  ```
  "chat_template_kwargs": {
      "enable_thinking": true,
      "clear_thinking": false
  }
  ```

- When using `vLLM` and `SGLang`, thinking mode is enabled by default when sending requests. If you want to disable the thinking switch, you need to add the `"chat_template_kwargs": {"enable_thinking": False}` parameter.

- Both support tool calling. Please use OpenAI-style tool description format for calls.

Citation

If you find our work useful in your research, please consider citing the following paper:

```
@misc{5team2025glm45agenticreasoningcoding,
      title={GLM-4.5: Agentic, Reasoning, and Coding (ARC) Foundation Models},
      author={GLM Team and Aohan Zeng and Xin Lv and Qinkai Zheng and Zhenyu Hou and Bin Chen and Chengxing Xie and Cunxiang Wang and Da Yin
        and Hao Zeng and Jiajie Zhang and Kedong Wang and Lucen Zhong and Mingdao Liu and Rui Lu and Shulin Cao and Xiaohan Zhang and
        Xuancheng Huang and Yao Wei and Yean Cheng and Yifan An and Yilin Niu and Yuanhao Wen and Yushi Bai and Zhengxiao Du and Zihan Wang
        and Zilin Zhu and Bohan Zhang and Bosi Wen and Bowen Wu and Bowen Xu and Can Huang and Casey Zhao and Changpeng Cai and Chao Yu and
        Chen Li and Chendi Ge and Chenghua Huang and Chenhui Zhang and Chenxi Xu and Chenzheng Zhu and Chuang Li and Congfeng Yin and
        Daoyan Lin and Dayong Yang and Dazhi Jiang and Ding Ai and Erle Zhu and Fei Wang and Gengzheng Pan and Guo Wang and Hailong Sun and
        Haitao Li and Haiyang Li and Haiyi Hu and Hanyu Zhang and Hao Peng and Hao Tai and Haoke Zhang and Haoran Wang and Haoyu Yang and
        He Liu and He Zhao and Hongwei Liu and Hongxi Yan and Huan Liu and Huilong Chen and Ji Li and Jiajing Zhao and Jiamin Ren and Jian
        Jiao and Jiani Zhao and Jianyang Yan and Jiaqi Wang and Jiayi Gui and Jiayue Zhao and Jie Liu and Jijie Li and Jing Li and Jing Lu
        and Jingsen Wang and Jingwei Yuan and Jingxuan Li and Jingzhao Du and Jinhua Du and Jinxin Liu and Junkai Zhi and Junli Gao and Ke
        Wang and Lekang Yang and Liang Xu and Lin Fan and Lindong Wu and Lintao Ding and Lu Wang and Man Zhang and Minghao Li and Minghuan
        Xu and Mingming Zhao and Mingshu Zhai and Pengfan Du and Qian Dong and Shangde Lei and Shangqing Tu and Shangtong Yang and Shaoyou
        Lu and Shijie Li and Shuang Li and Shuang-Li and Shuxun Yang and Sibo Yi and Tianshu Yu and Wei Tian and Weihan Wang and Wenbo Yu
        and Weng Lam Tam and Wenjie Liang and Wentao Liu and Xiao Wang and Xiaohan Jia and Xiaotao Gu and Xiaoying Ling and Xin Wang and
        Xing Fan and Xingru Pan and Xinyuan Zhang and Xinze Zhang and Xiuqing Fu and Xunkai Zhang and Yabo Xu and Yandong Wu and Yida Lu
        and Yidong Wang and Yilin Zhou and Yiming Pan and Ying Zhang and Yingli Wang and Yingru Li and Yinpei Su and Yipeng Geng and Yitong
        Zhu and Yongkun Yang and Yuhang Li and Yuhao Wu and Yujiang Li and Yunan Liu and Yunqing Wang and Yuntao Li and Yuxuan Zhang and
        Zezhen Liu and Zhen Yang and Zhengda Zhou and Zhongpei Qiao and Zhuoer Feng and Zhuorui Liu and Zichen Zhang and Zihan Wang and
        Zijun Yao and Zikang Wang and Ziqiang Liu and Ziwei Chai and Zixuan Li and Zuodong Zhao and Wenguang Chen and Jidong Zhai and Bin
        Xu and Minlie Huang and Hongning Wang and Juanzi Li and Yuxiao Dong and Jie Tang},
      year={2025},
      eprint={2508.06471},
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      archivePrefix={arXiv},
      primaryClass={cs.CL},
      url={https://arxiv.org/abs/2508.06471},
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